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Time domain topological gradient and time reversal analogy: an inverse method for ultrasonic target detection

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Abstract

To find the shape and position of one or several obstacles embedded in a 2D or 3D elastic medium given ultrasonic measurements performed with a linear array of sensors, we use a topological gradient method. It consists in minimizing a cost function which evaluates the adequation between the measurements obtained for the inspected medium and measurements performed on a reference medium known as safe. The topological gradient is a function of both the solutions of the forward and adjoint problems. The mathematical solution of such an adjoint problem corresponds to a physical time reversal operation. This inverse method is therefore physically justified and inheritance of the refocusing properties of the time reversal phenomenon is expected.
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1. Introduction

Localization and characterization of defects or obstacles in a fluid or solid elastic medium is of first importance in various scientific fields such as non-destructive testing, medical imaging, geophysics or submarine detection. We are particularly interested in the non-destructive testing application where data like the position and the nature of the objects are wanted, given a minimum number of measurements.

This inverse problem has been widely studied and a significant number of methods have been imagined to solve it. However to our knowledge the problem is still open since no method leads to a consensus.

Among the available literature, the works of Prada et al. [1–3] gave rise to the now famous DORT method, in great relation with the physical concept of time reversal mirror that has been widely studied and experienced by

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Fink et al. [4,5]. DORT (a French acronym for Decomposition of the Time Reversal Operator) relies on an algebraic analysis of the time reversal operator that is the multiplication of the matrix resulting from the acquisition through the scattering medium with its complex conjugate. Significant eigenvalues give the number of scatterers while the associated eigenvectors give their locations.

Following these pioneering works, Borcea et al. [6] and Blomgren et al. [7] on the one hand, Devaney et al. on the other hand [8,9] have studied the principle of time reversal to solve the inverse problem in random multiple scattering media. The resulting imaging methods are based on the computation of an objective functional whose maxima are an estimation of the position of the scattering objects. Resulting methods as MUSIC [8,9] and SAT [6] are examples that can be found in literature.

Geophysics also brought elegant techniques in inverse problems. In particular the theory of migration imagined by Tarantola [10] can be used to find the elastic stiffness or density of obstacles in soils, including the attenuation of the propagation medium. The problem is seen as a parameter optimization problem in a functional space. Thus iterative gradient methods are used, each iteration consisting of the propagation of the actual source in the medium and the propagation of the residuals (the difference between the measurements and the reference solution) acting as sources backwards in time. We will see that this latter approach has, to some extent, some similarities with the *topological gradient* method presented in this paper.

The topological gradient approach comes from the mathematical community of shape optimization. This method has been originated by Schumacher[11,12], partially generalized by Sokolowski [13] and then developed and extended by Masmoudi et al. [14,15] to electromagnetics and linear elasticity. We present here an extension to the elastodynamic case in order to mimic the non-destructive testing conditions. The principle is the following. Let us assume that we have a reference medium, known as safe or exempt of some kind of defects, and that we want to inspect an 'unknown' medium of the same type as the reference one, but which may contain defects or undesirable inhomogeneities. The idea is to write a cost function which evaluates the 'distance' between the measurements obtained for the unknown medium and the measurements performed on the reference medium. Starting from an initial design (the reference medium), a sensitivity analysis of the cost function with respect to the introduction of infinitesimal 'holes' in this domain is done. The result of this analysis governs the modification of the topology of the medium. The new virtually modified medium is expected to converge towards the topology of the inspected one. The sensitivity analysis is performed by writing an asymptotic expansion of the cost function with respect to the topology of the medium (that is why the method is also called topological asymptotic). The first order terms of the asymptotic expansion give the expression of a 'topological gradient' whose more negative values indicate where to insert holes to decrease the value of the cost function. It is worth noting that the word 'hole' is here a generic expression that designates an embedded zone with an abrupt contrast of elastic properties with respect to the rest of the medium. Until now, no other proof than a convergence towards a local minimum of the cost function exists.

The aim of this article is two-fold. First we will present the time domain topological gradient method (TDTG) and associated results in the field of non-destructive testing. Second, we would like to highlight that this mathematical optimization process is intimately linked to the physics of time reversal and is then physically justified by the natural refocusing properties of time reversal. We do not pretend to give a rigorous mathematical demonstration of the convergence towards the global minimum (the true topology of the inspected medium) but we give physical arguments that give confidence into the results obtained with the TDTG method.

The analogy between the TDTG method, the migration method and the time reversal comes from the fact that the computation of the gradient imposes to solve an adjoint problem which is found to be a time reversal operation. This propagation-backpropagation process also recalls the well known ART method in x-ray tomography and the method imagined by Natterer at fixed frequency [16]. Other works on the imaging of elastic stiffness with an adjoint method have also been worked out in the frequency domain by Oberai et al. [17,18].

In Section 2 we give the basic equations of the elastodynamic problem. The principle of the topological gradient method is presented in Section 3. In Sections 4 and 5, the extension to the elastodynamic case and the algorithm of the

TDTG method are presented. The numerical results of Section 6 show the efficiency of the method in various non-destructive testing configurations while Section 7 is devoted to the analogy with time reversal and its consequences in terms of justification of the TDTG method as well as of the expected inheritance of physical properties.

2. The elastodynamic problem

Let us assume a reference medium Ω_0, known as safe. The problem is to be able to localize undesirable heterogeneities in a medium Ω_{insp} by inspecting it with a linear array of transducers.

Let Γ be the external frontiers of $\Omega \in \mathbb{R}^N$, $N = 2$ or 3. The measurements can be done on a part Γ_m of $\bar{\Omega} = \Omega \cap \Gamma$. The topological dimension of Γ_m can be less than N.

The conservation of energy and Hooke's law yield the equations of ultrasonic propagation in elastic media

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \operatorname{div} \sigma(\mathbf{u}) + \mathbf{f}, \qquad \sigma(\mathbf{u}) = \mathbf{C}\varepsilon(\mathbf{u}), \tag{1}$$

where $\mathbf{u}$ is the displacement vector, σ and $\varepsilon(\mathbf{u})$ are respectively the stress and strain tensors, $\mathbf{C}$ is Hooke's tensor, and $\mathbf{f}$ is a source of ultrasonic waves (a force). The stress tensor can be written as a function of the displacement vector: $\varepsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^{\mathrm{T}})$, then we can put (1) into the general form

$$\frac{\partial^2 u}{\partial t^2} + Au = \mathbf{F}, \tag{2}$$

where A is a linear operator. A is self-adjoint if there is no intrinsic attenuation in the medium ($A^* = A$).

3. The topological gradient method

In this section we introduce the major notions and notations of the topological asymptotic method. We do not pretend to give full mathematical demonstrations of the results presented, but only to give the reader the matter to understand the following of this article. The interested reader can consult references given in bibliography [15,19] for more mathematical details.

Unlike in the forward problem where the unknown is the field created in the domain by the ultrasonic source, the unknown of our inverse problem is the topology of the inspected domain Ω_{insp}. We recall that in the following the word 'hole' designates any kind of zone having an abrupt rupture of elastic properties (or density) with respect to the rest of the medium.

3.1. The topological optimization process

The process is initialized with an initial guess which coincides with the reference domain Ω_0. We want to modify the topology of this initial design by inserting 'holes' to make it tend towards the topology of the inspected medium Ω_{insp}. The question is where must we insert those holes?

Let ω be such a 'hole', centered on 0, and $\partial\omega$ its boundary. We call $\omega_{x,\epsilon}$ a 'hole' of radius ϵ, centered on $x \in \Omega_0 : \omega_{x,\epsilon} = \epsilon\omega + x$. The domain perforated by $\omega_{x,\epsilon}$ is $\Omega_\epsilon = \Omega_0 \backslash \overline{\omega_{x,\epsilon}}$. The principle of the topological asymptotic method is depicted in Fig. 1.

One writes a cost function j (estimating the 'distance' between the reference medium and the unknown medium) and minimizes it by computing an asymptotic expansion of j about the reference medium. Considering the boundary conditions on the edge of the hole (defining what kind of 'hole' it is), one can compute an asymptotic expansion at

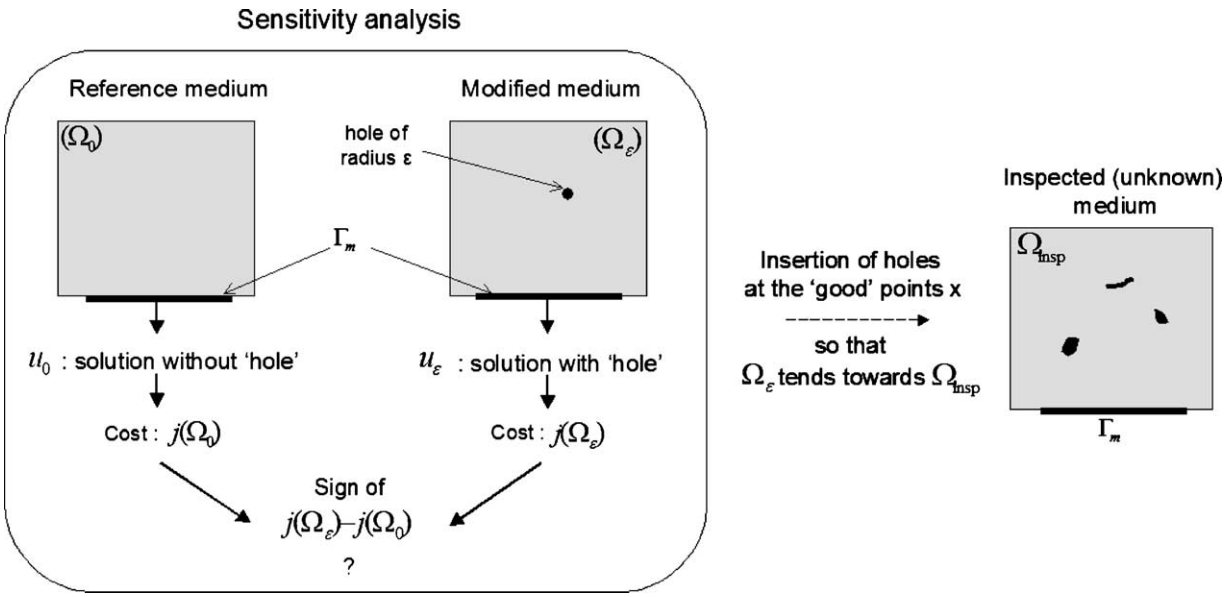

Fig. 1. Principle of the topological asymptotic.

the first order of a cost functional j under the form

$$j(\Omega_\epsilon) = j(\Omega_0) + f(\epsilon)g(x) + \mathrm{o}(f(\epsilon)) \tag{3}$$

with $\lim_{\epsilon\to 0} f(\epsilon) = 0$ and $f(\epsilon) > 0$.

The functional g gives, at every point x of Ω_0, a sensitivity analysis of the problem as to the insertion of a hole $\omega_{x,\epsilon}$ at that point. Indeed, since f is positive, the more g is negative at x the more the insertion of a hole at that point decreases the cost function. g is called the *topological gradient* in reference to the terminology of classical optimization processes. In other words, the insertion of a hole at x such that $g(x) < 0$ is a descent direction in the topological optimization process.

3.2. The cost function

The cost function depends on the domain Ω via the solution u_Ω of the equations of elastodynamics (1) on Ω

$$j(\Omega) = J(u_\Omega).$$

Since the measurements are performed only on a part (or a trace) Γ_m of Ω, we only have access to the solution u_Ω on Γ_m, and a natural cost function is

$$j(\Omega) = \frac{1}{2}\int_{\mathbb{R}}\left(\int_{\Gamma_m} |u_\Omega - u_m|^2 \,\mathrm{d}x\right)\mathrm{d}t. \tag{4}$$

In particular, $j(\Omega_0) = \frac{1}{2}\int_{\mathbb{R}}(\int_{\Gamma_m} |u_0 - u_m|^2 \,\mathrm{d}x)\,\mathrm{d}t$ evaluates the adequation, on the location Γ_m of the measurements and over the entire acquisition time, between the solution u_0 of the problem in the reference domain Ω_0 and the solution u_m of the problem in the unknown domain Ω_{insp}.

Table 1
Topological asymptotic for linear elasticity

Boundary condition on $\partial\omega_\epsilon$	$f(\epsilon)$	Topological gradient: $g(x)$
2D Dirichlet	$-\dfrac{2\pi}{\log(\epsilon)}$	$\dfrac{2\mu(\mu+\eta)}{2\mu+\eta}u_0 \cdot v_0$
3D Dirichlet	$4\pi\epsilon$	$\dfrac{3\mu(\lambda+2\mu)}{2\lambda+5\mu}u_0 \cdot v_0$
2D Neumann	$\pi\epsilon^2$	$-\dfrac{\mu+\eta}{2\mu\eta}(4\mu\sigma(u_0):\varepsilon(v_0)+(\eta-2\mu)\operatorname{tr}\sigma(u_0)\operatorname{tr}\varepsilon(v_0))$
3D Neumann	$\dfrac{4}{3}\pi\epsilon^3$	$-\dfrac{3(\lambda+2\mu)}{4\mu(9\lambda+14\mu)}(20\mu\sigma(u_0):\varepsilon(v_0)+(3\lambda-2\mu)\operatorname{tr}\sigma(u_0)\operatorname{tr}\varepsilon(v_0))$

3.3. The topological asymptotic for linear elasticity

Here we give the formulas of the topological gradient in the case of linear elasticity. This work has been done by Garreau et al. in [15]. Using these expressions we will generalize the topological asymptotic to the case of elastodynamics in Section 4.

The type of boundary conditions on the edge of the 'holes' has a central role when establishing the topological asymptotic expansion. In the case of defect investigations with ultrasound in acoustic or elastic media, three types of boundary conditions must be considered:

- The Dirichlet condition corresponds to the insertion of an infinitely rigid ball and imposes the displacements at the interface medium/ball to be zero.
- The Neumann condition models a cavity (a real hole) and imposes the normal stress at the boundary of the hole to be zero.
- The impedance type boundary condition corresponds to an elastic inclusion. It is not treated in this article but will be the subject of a future work.

Considering the classical linear elasticity problem

$$\begin{cases} -\operatorname{div}\sigma(u)=0 & \text{in } \Omega \\ \sigma(u)\mathbf{n}=F & \text{on } \Gamma_{\text{Neumann}} \\ u=0 & \text{on } \Gamma_{\text{Dirichlet}} \end{cases} \tag{5}$$

the topological asymptotic is as given in Table 1 [15].

The quantities $f(\epsilon)$ and $g(x)$ are those defined in (3), and

- u_0 and $\sigma(u_0)$ are respectively the displacement and stress fields solution of the forward problem solved in the reference medium Ω_0.
- v_0 and $\varepsilon(v_0)$ are respectively the displacement and strain fields solution of the adjoint problem solved in the reference medium Ω_0.
- λ and μ are the Lamé coefficients of Ω_0.
- For the 2D cases,

$$\begin{cases} \eta=\dfrac{\mu(3\lambda+2\mu)}{\lambda+2\mu} & \text{in plane stress} \\ \eta=\lambda+\mu & \text{in plane strain} \end{cases} \tag{6}$$

- ϵ is the radius of the 'hole'.

Remark. In any case the topological gradient g involves both the solutions of a forward and an adjoint problem in the reference medium Ω_0. The need to solve an adjoint problem is shown in Appendix A. The case of this adjoint problem will be discussed later in this article. In particular we will see that it can be physically interpreted when considered in time domain.

4. The time domain topological gradient for elastodynamics

The extension of the formulas of Table 1 to the time domain dynamic problem are presented in this section. The key is to use the time-harmonic case and then the duality of the frequency and time domains through the Fourier transform.

The time-harmonic problem is

$$\rho\omega^2 u + \operatorname{div}\sigma(u) = 0.$$

Let us call $J^\omega(\hat{u}(\cdot,\omega))$ the cost functional of the time-harmonic problem, where $\hat{u}(x,\omega) = \int_{\mathbb{R}} u(x,t)\,\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t$ is the Fourier transform of $u(x,t)$. Starting with the cost functional of the time domain problem (4) and using successively Fubbini's theorem and Parseval's equality, it comes

$$J(u) = \frac{1}{2}\int_{\mathbb{R}}\left(\int_{\Gamma_m} |u-u_m|^2\,\mathrm{d}x\right)\mathrm{d}t = \int_{\mathbb{R}}\left(\frac{1}{2}\int_{\Gamma_m} |\hat{u}-\hat{u}_m|^2\,\mathrm{d}x\right)\mathrm{d}\omega = \int_{\mathbb{R}} J^\omega(\hat{u}(\cdot,\omega))\,\mathrm{d}\omega. \tag{7}$$

Yet, at a given frequency, the asymptotic expansion for $J^\omega(\hat{u})$ is known from the one of linear elasticity. For instance for 2D Neumann:

$$J^\omega(\hat{u}_\epsilon) - J^\omega(\hat{u}_0) = \pi\epsilon^2\left(-\frac{\mu+\eta}{2\mu\eta}\left(4\mu\sigma(\hat{u}_0):\varepsilon(\hat{v}_0) + (\eta-2\mu)\operatorname{tr}\sigma(\hat{u}_0)\operatorname{tr}\varepsilon(\hat{v}_0)\right) + \rho\omega^2\hat{u}_0\cdot\hat{v}_0\right) + \mathrm{o}(\epsilon^2).$$

Then using (7) and Parseval's equality, one obtains the expressions for the time domain problem by writing:

$$\begin{aligned} J(u_\epsilon) - J(u_0) &= \int_{\mathbb{R}} (J^\omega(\hat{u}_\epsilon) - J^\omega(\hat{u}_0))\,\mathrm{d}\omega \\ &= \pi\epsilon^2\int_{\mathbb{R}}\left(-\frac{\mu+\eta}{2\mu\eta}(4\mu\sigma(\hat{u}_0):\varepsilon(\hat{v}_0) + (\eta-2\mu)\operatorname{tr}\sigma(\hat{u}_0)\operatorname{tr}\varepsilon(\hat{v}_0)) + \rho\omega^2\hat{u}_0\cdot\hat{v}_0\right)\mathrm{d}\omega + \mathrm{o}(\epsilon^2) \\ &= \pi\epsilon^2\int_{\mathbb{R}}\left(-\frac{\mu+\eta}{2\mu\eta}(4\mu\sigma(u_0):\varepsilon(v_0) + (\eta-2\mu)\operatorname{tr}\sigma(u_0)\operatorname{tr}\varepsilon(v_0)) - \rho\partial_t u_0\cdot\partial_t v_0\right)\mathrm{d}t + \mathrm{o}(\epsilon^2). \end{aligned} \tag{8}$$

It has been shown [20] that for the Dirichlet boundary condition cases, the terms in $(\rho\partial_t u_0\cdot\partial_t v_0)$ are of higher order in ϵ and therefore do not appear in the first order asymptotic expansion yielding the topological gradient. Table 2 shows the formulas for the topological gradient for elastodynamics depending on the boundary condition at the edge $\partial\omega_\epsilon$ of the 'holes' and of the space dimension of the problem.

What is important to notice in these expressions of the topological gradient is that the dominant terms are, roughly speaking, scalar products of the solutions of the forward and adjoint problems at every point of the computation domain, and integrated over the acquisition time. In particular rigid ball inclusions involve the scalar product of the displacements, while cavities involve the tensorial product of the stress tensor of the forward problem with the strain tensor of the adjoint problem.

Practically we will not have access to the solutions for $t \in \mathbb{R}^+$, but only over an interval $[0, T]$. Then T must be taken large enough so that the amplitude of the fields in the computation domain after the time T is weak enough to be neglected when computing the topological gradient.

Table 2
Topological asymptotic for elastodynamics

Boundary condition on $\partial\omega_\epsilon$	$f(\epsilon)$	Topological gradient: $g(x)$
2D Dirichlet	$-\dfrac{2\pi}{\log(\epsilon)}$	$\int_{\mathbb{R}^+}\left(\dfrac{2\mu(\mu+\eta)}{2\mu+\eta}u_0\cdot v_0\right)\mathrm{d}t$
3D Dirichlet	$4\pi\epsilon$	$\int_{\mathbb{R}^+}\left(\dfrac{3\mu(\lambda+2\mu)}{2\lambda+5\mu}u_0\cdot v_0\right)\mathrm{d}t$
2D Neumann	$\pi\epsilon^2$	$\int_{\mathbb{R}^+}\left(-\dfrac{\mu+\eta}{2\mu\eta}(4\mu\sigma(u_0):\varepsilon(v_0)+(\eta-2\mu)\operatorname{tr}\sigma(u_0)\operatorname{tr}\varepsilon(v_0))-\rho\partial_t u_0\cdot\partial_t v_0\right)\mathrm{d}t$
3D Neumann	$\dfrac{4}{3}\pi\epsilon^3$	$\int_{\mathbb{R}^+}\left(-\dfrac{3\pi(\lambda+2\mu)}{4\mu(9\lambda+14\mu)}(20\mu\sigma(u_0):\varepsilon(v_0)+(3\lambda-2\mu)\operatorname{tr}\sigma(u_0)\operatorname{tr}\varepsilon(v_0))-\rho\partial_t u_0\cdot\partial_t v_0\right)\mathrm{d}t$

5. Implementation of the TDTG method

In this section we present the algorithm of the time domain topological gradient. Before all, we have seen in the previous sections that we need to solve an adjoint problem. The fact is that we have a forward solver, but not necessarily an adjoint solver. So how can we solve this adjoint problem?

5.1. Solving the adjoint problem

5.1.1. Time-harmonic problem

In the time-harmonic regime, the forward problem can be written as (Appendix A, Eq. A.1):

$$Au = f$$

and the associated adjoint problem is (Appendix A, Eq. A.12)

$$A^*v = -D_u J(u_0), \tag{9}$$

where u_0 is the solution to the forward problem, $D_u J$ designates the derivative of J with respect to the variable u, and A^* is the adjoint operator of A. In both cases of Laplacian and elastodynamics, this operator is self-adjoint ($A^* = A$). The adjoint problem can then be solved with the forward solver, the only difference being the source. This source term is found by writing the weak formulation (see Appendix A), differentiating the cost function then coming back to the strong formulation. The time-harmonic adjoint problem is then

$$A^*v = -\overline{(u_0 - u_m)}\delta_{\Gamma_m}, \tag{10}$$

where $\overline{(u_0 - u_m)}$ is the complex conjugate of $(u_0 - u_m)$ and δ_{Γ_m} is the Kronecker delta function for Γ_m ($\delta_{\Gamma_m} = 1$ on Γ_m and 0 anywhere else).

5.1.2. Time domain problem

Solving the adjoint problem in the time domain is a little trickier than in the time-harmonic case since if the forward problem is (2):

$$\frac{\partial^2 u}{\partial t^2} + Au = F \quad \forall t \in [0, T]$$

with the initial condition

$$u(x, 0) = 0 \tag{11}$$

then the adjoint problem writes

$$\partial_t^2 v + A^* v = -D_u J(u_0) \quad \forall t \in [T, 0], \qquad v(x, T) = 0. \tag{12}$$

Indeed Lions [21] and Tarantola [10] have shown that the adjoint operator of $(\partial_t^2 + A)$ is $(\partial_t^2 + A^*)$ and that the initial condition of the associated adjoint problem occurs at $t = T$. One can show in the same manner as in the appendix but considering the time domain problem instead of the time-harmonic one that our adjoint problem writes as (12). It is worth noting that the initial condition of the adjoint problem occurs at $t = T$. We get round this trick by writing $t = T - t$, that is to say that we solve the problem backwards in time from T to 0. If there is no intrinsic attenuation in the medium, A is self-adjoint and (12) becomes:

$$(\partial_t^2 v + Av)(x, T - t) = -D_u J(u_0)(x, T - t) \quad \forall t \in [0, T], \qquad v(x, T) = 0. \tag{13}$$

Then the adjoint problem in the time domain is identical to the forward problem, except that the time travels in the opposite direction ! Consequently the adjoint problem can be solved by the forward solver provided that the source term is time reversed.

After computing this source term, we have

$$\begin{aligned} &(\partial_t^2 v + Av)(x, T - t) = 0 \quad \forall x \in \Omega \backslash \overline{\Gamma}_m \ \forall t \in [0, T], \\ &(\partial_t^2 v + Av)(x, T - t) = -(u_0 - u_m)(x, T - t) \quad \forall x \in \Gamma_m \ \forall t \in [0, T], \qquad v(x, T) = 0, \end{aligned} \tag{14}$$

where u_0 is the solution of the forward problem in the reference medium Ω_0 and u_m is the result of the measurement done on the inspected medium Ω_{insp}.

Remark. The adjoint problem of the time-harmonic case corresponds to phase conjugation. The classical duality between time reversal and phase conjugation is then well verified through the adjoint problem.

5.2. Algorithm of the TDTG method

We can now give the algorithm of the time domain topological gradient method.

1. *Measurement on the unknown medium* Ω_{insp} with a linear array of ultrasonic transducers.
   - We get $u_m(x, t)$ on Γ_m, $x \in \Gamma_m$ and $t \in [0, T]$.
2. *Numerical solving of the forward problem in the reference medium* Ω_0.
   - We record $u_0(x, t)$ on Ω_0 (in particular on Γ_m) and for all time step in $[0, T]$.
3. *Numerical solving of the adjoint problem in the reference medium* Ω_0 with the forward solver. The source of this problem is $(u_m - u_0)(T - t)$, for $t \in [0, T]$.
   - We record $v_0(x, T - t)$ on Ω_0, and for all time step in $[0, T]$.
     Notice: The output v_0 of this solving is time reversed, i.e. when $t = 0$ one have $v_0(T)$, and when $t = T$ one have $v_0(0)$. We must put back this solution in the direction of increasing time before using it in the computation of the topological gradient g.
4. *Computation of the topological gradient* g following the formulas of Table 2 and plotting of the isolevels of g.

Now we have to decide whether the isolevels of the topological gradient give a sufficient estimation of the topology of the medium or not. If not, it is possible to refine this estimation using the fixed point algorithm that follows. The measurement on the unknown medium (step 1 of the algorithm above) is done once for all. Let Ω_1 be the result of the algorithm above.

- *Initialization* with Ω_1.
- $k = 0$.
- *While* $\Omega_{k+1} \neq \Omega_k$

  1. $k = k + 1$,
  2. compute u_k the forward solution in the domain Ω_k,
  3. compute v_k the adjoint solution in the domain Ω_k,
  4. compute g the topological gradient,
  5. update the domain Ω_{k+1}: modify the state of the α_k% more negative points.

- *Loop*.

The parameters α_k rule the speed of convergence. High values of α will give a quick but poorly accurate solution while small values of α will yield a slow but very accurate solution.

Anyway we will see in Section 6 that even when a precise solution requires the iterating process, we can have a quite good idea of the solution without iterating using an approximate expression of the topological gradient. This makes the TDTG method very efficient in terms of computation time since only two forward problems need to be solved (one forward and one adjoint), while traditional optimization processes would require many more solving to converge. In counterpart we must admit that the major drawback of this method is its memory cost. Indeed it is preferable to use a computer with a large enough memory to stock the result of a whole solving, that means the concerned fields at every point of the computation domain and at every time step. One can get round this drawback by organizing a periodic saving on the hard disk to avoid the swapping at every type step which would be dramatic for the computation time.

5.3. *The forward solver*

It is now clear that the TDTG method only needs one solver for both the adjoint and forward problems. Let us previously notice that any kind of solver of the equations of acoustics or elastodynamics (depending on the considered problem) can be used, provided it is a time domain solver. The underlying method can be either numerical (finite differences, finite elements, boundary elements, spectral elements) or semi-analytical (ray pathing and refinements).

5.3.1. *Numerical method*

We have developed a finite difference C++ code following Virieux numerical scheme [22] which is accurate at the order 2 in space and time and intrinsically centered. It allows one to take into account abrupt ruptures of elastic properties or density such as fluid/solid interfaces. This code is now integrated to the software ACEL developed by Tanter [23] and which is dedicated to the simulation of acoustic and elastic wave propagation.

5.3.2. *Boundary conditions*

The boundary conditions at the edges of the computation domain are either of the classical Dirichlet and Neumann type, or of absorbing type to simulate unbounded propagation. The implemented absorbing conditions are Perfectly Matched Layers following Collino and Tsogka [24].

6. Numerical results

In this section we present numerical results relative to non-destructive testing obtained with the TDTG method. The measurement step is up to now replaced by a numerical solving of the forward problem in the presence of obstacles. However, in order to be as close as possible to experimental conditions, only one component of the

displacement field (the one orthogonal to the array) is emitted and recorded by the transducers. In the first example we present results of the iterative method and introduce a quick imaging estimator derived from the topological gradient expression. The second example shows the great possibilities of this estimator. In a last result, we add noise to the measured data on a typical non-destructive testing configuration and we do not find dramatic degradations of the results. A future work will consider experimental data. The presented results are 2D since the 3D code is still being developed.

6.1. Defects in an isotropic solid

This first result is obtained following the iterative procedure of Section 5.2. We also show that the computation of a derived quantity of the topological gradient can give a good idea of the distribution of the defects from the first iteration.

The considered medium is an isotropic aluminium slab of density $\rho = 2572\,\mathrm{kg\,m^{-3}}$, the speeds of propagation are $v_\mathrm{p} = 6408\,\mathrm{m\,s^{-1}}$ and $v_\mathrm{s} = 3228\,\mathrm{m\,s^{-1}}$. The inspection is done in contact with the medium. The ultrasonic linear array is placed at the bottom of the slab. We use a 55 sensors array, all of them being used in emission and receive. Absorbing conditions are positioned at the boundaries of the computation domain, except at the bottom where a Dirichlet condition models the presence of the sensors. The emitted signal is a pulse of 1 μs at the central frequency of 2 MHz with a Gaussian wide frequency band spectrum. The defects are shown in Fig. 2(a), it corresponds to infinitely rigid cylinders whose sizes are respectively of the order of the compressional wavelength and sub-wavelength ($\lambda/4$).

6.1.1. The iterative process

Fig. 2(b) shows that the topological gradient is oscillating around the locus of the defects. The more negative values are situated at the bottom of the obstacles (Fig. 2 (c)) because the emitting/receiving transducer array is placed at the bottom of the medium. Then we can have a good description of the bottom of the defects, but cannot expect to be accurate in the shadow zones.

Fig. 2(c)–(e) presents the topology of the 'virtually' modified media Ω_k at successive iterations of the algorithm of Section 5.2. The number of selected points (in relation with the parameters α_k of Section 5.2) decreases as k grows. As expected, one observes that iteration after iteration the topology of Ω_k gets closer to the one of the inspected medium (at least on the insonified side). After eight iterations the algorithm has converged in the sense that $\Omega_9 = \Omega_8$. As a verification we plot in Fig. 3 the recorded B-scans for the inspected medium and for the medium Ω_8. We also plot the difference of these two B-scans. A B-scan is a 2D representation of the signals recorded by a linear array. The ith column is the temporal signal (in number of steps) recorded on the transducer i of the array. The plot of the difference of the B-scans is normalized to the maximum value of the echo signals recorded on the inspected medium.

The value of the cost function (4) at the 8th iteration is the energy contained in the B-scan window of Fig. 3(c), the initial value of this cost function being the energy contained in the B-scan window of Fig. 3(a). Fig. 2(f) shows the decreasing of the cost function along the iterative process, normalized to the value at iteration 0. As regards to the so-positioned linear array, the inspected medium (Fig. 2(a)) and Ω_8 (Fig. 2(e)) are equivalent since the echoes generated by those two configurations are almost identical. The bottom of the obstacles being accurately reproduced, we understand why the algorithm stops at the 8th iteration although Ω_8 is still far from the goal (the inspected medium). An accurate determination of the shape of the whole defects could be achieved by displacing the array on the four sides of the medium.

6.1.2. A quick imaging estimator

In order to have a quicker and more representative result (although less rigorous and precise) we can compute a simple expression derived from the topological gradient and involving only the results obtained without iterating the process. This quantity g_0, that we call *topological energy*, is in fact a product of the energies of the forward and

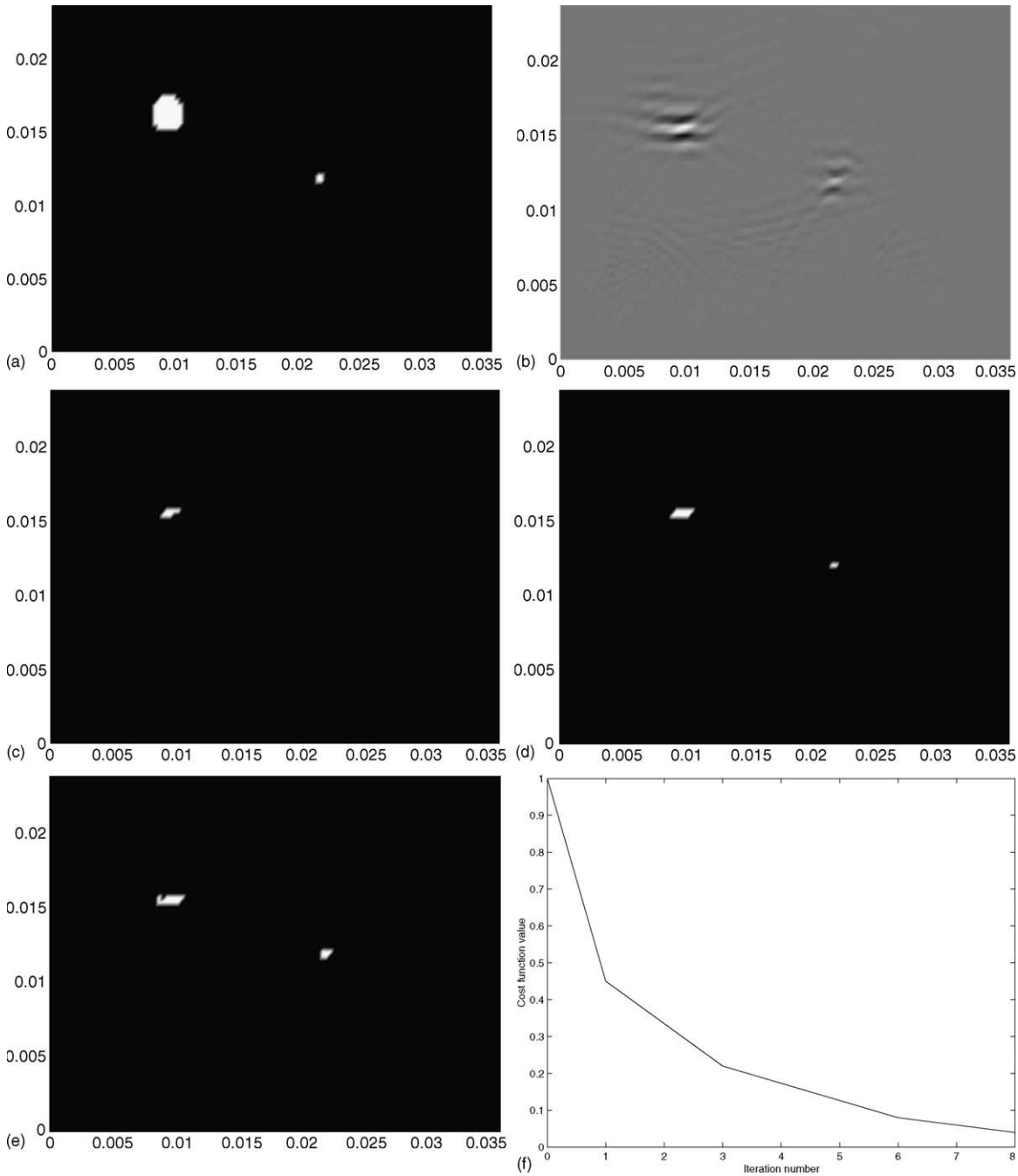

Fig. 2. Imaging of two defects of different sizes (λ and $\lambda/4$) with the iterative procedure: (a) positions of the defects, (b) iteration 1: isolevels of the topological gradient, (c) iteration 1: Ω_1, (d) iteration 3: Ω_3, (e) iteration 8: Ω_8 and (f) evolution of the cost function.

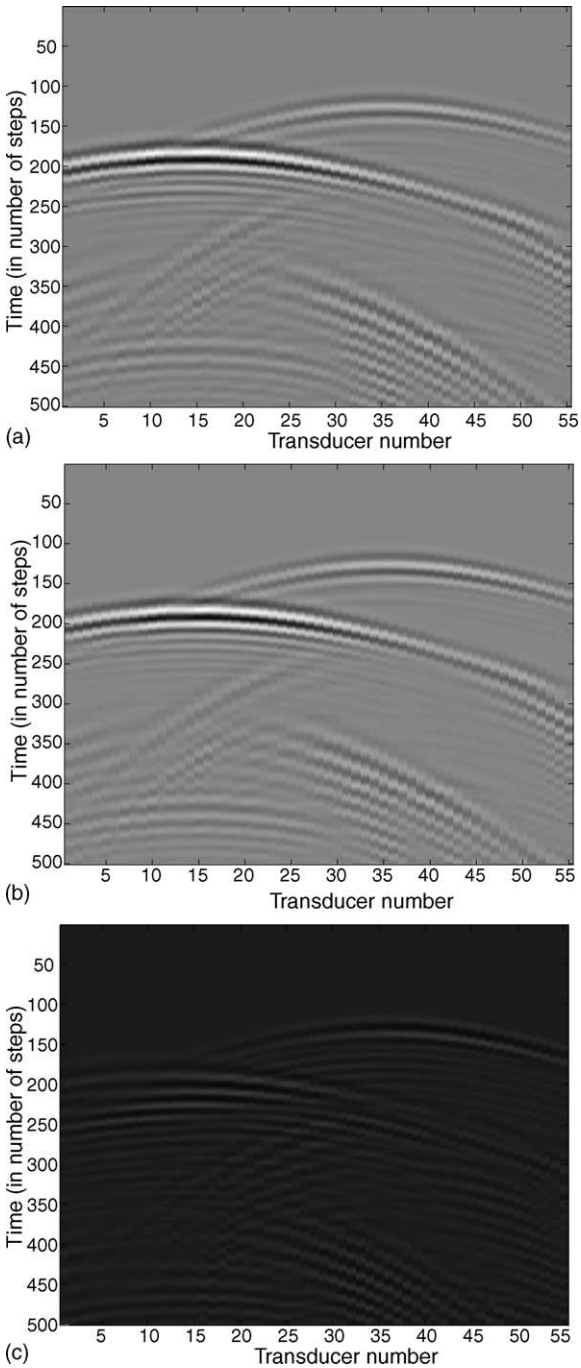

Fig. 3. Recorded signals on the linear array: (a) for the inspected medium, (b) for the medium Ω_8 and (c) difference of these signals.

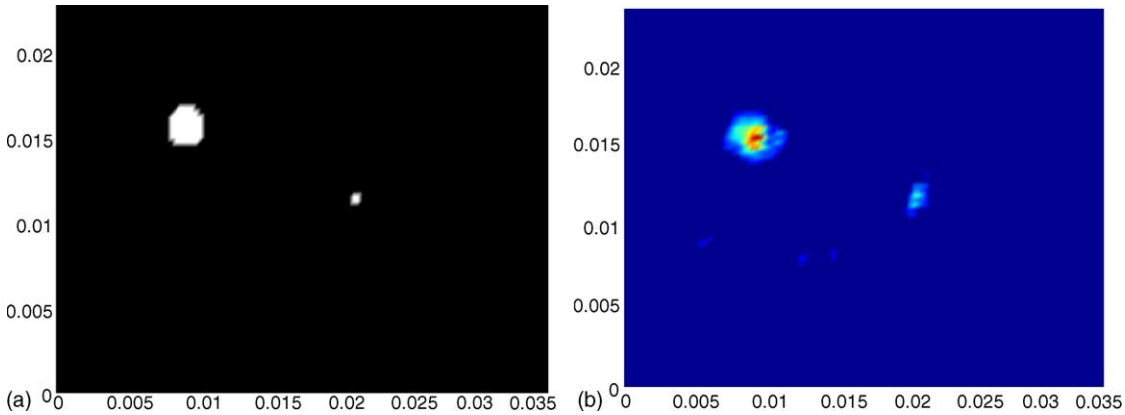

Fig. 4. Imaging of two defects of different sizes (λ and $\lambda/4$) with the quantity g_0: (a) positions of the defects and (b) isolevels of the topological energy g_0.

adjoint states, integrated in time

$$g_0(x) = \int_0^T ||U_0(x,t)||^2 \, ||V_0(x,t)||^2 \, \mathrm{d}t, \tag{15}$$

where U_0 and V_0 are respectively the displacement fields solutions of the forward and adjoint problems in the reference medium at the first iteration of the process. $||U||$ denotes the Euclidean norm of the vector U. This quantity g_0 removes the natural oscillations of the topological gradient.

One observes in Fig. 4 that the isolevels of g_0 give a good idea of the sizes and positions of the defects in the medium. The reason why g_0 yields satisfying results will become clear in Section 7 through the visual example of Fig. 8 and the analogy with time reversal.

The computation of g_0 is straightforward and gives an approximate but often sufficient idea of the distribution of the defects from the non-iterated results. The method is then very few time consuming. However we observe that the resolution of the smallest defect is poor. The location of the defect is correct but the spot of the topological energy g_0 is much more larger than the real size of this defect. This is both due to the limited aperture of the array and to the imperfect refocusing at the locus of the obstacle which are classical limitations for refocusing with time reversal mirrors (TRM) as explained in Section 7.

6.2. Multiple shaped defects

Here we put three defects of various shapes in the aluminium slab. These defects are well resolved since they are separated from more than a wavelength. We use the same linear array as before. The result of the topological energy g_0 is shown in Fig. 5.

The three defects are detected and localized. The shapes are relatively similar to the ones of the obstacles, except in the shadow zones.

6.3. Defect in an immersed slab

Non-destructive evaluation of materials such as composites is very often done by immersing the slab in water to diminish the entrance impedance contrast. We show here an example of a carbon-epoxy material (homogenized), with one defect of the order of the compressional wavelength in the material, as shown in Fig. 6. The characteristics

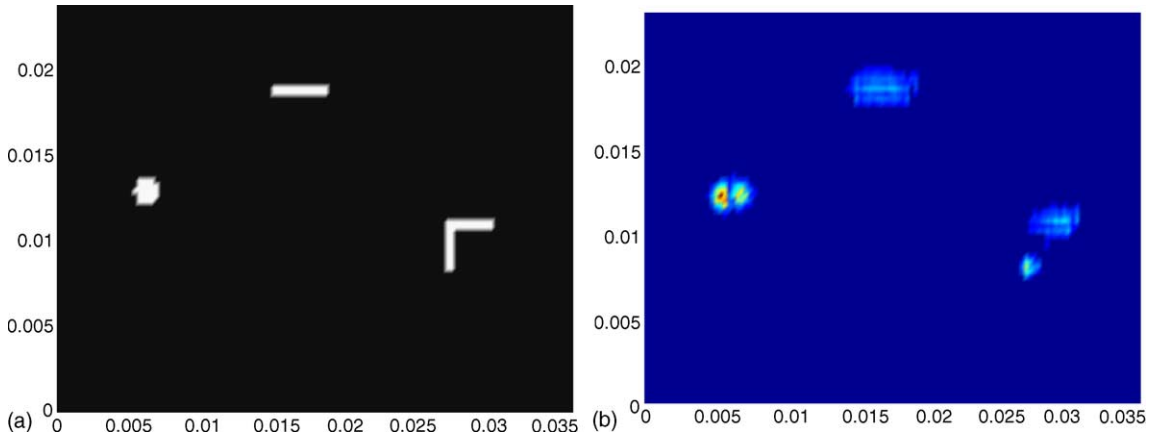

Fig. 5. Imaging of multiple shaped defects: (a) positions of the defects and (b) isolevels of the topological energy g_0.

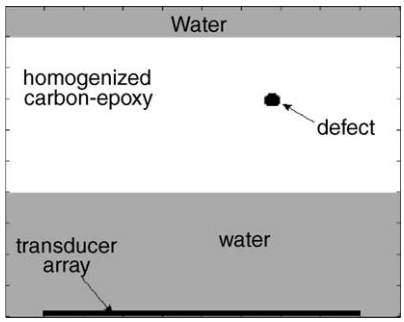

Fig. 6. Homogenized carbon-epoxy slab immersed in the water.

of the homogenized material are: $\rho = 1550\,\mathrm{kg\,m^{-3}}$, $v_\mathrm{p} = 3000\,\mathrm{m\,s^{-1}}$ and $v_\mathrm{s} = 900\,\mathrm{m\,s^{-1}}$. The array of transducers is placed as depicted in Fig. 6. The input signal is the same as before.

We successively use linear arrays of 36 transducers of 0.25 mm each ($\lambda_\mathrm{water}/2$) and of 18 transducers of 0.5 mm each. We add noise to the simulated measurements to mimic the experimental measurement noise. This is a white noise of 5% of the maximum value of the emitted pulse corresponding to a signal to noise ratio of 11 on the recorded signal including the entry echo. The topological gradient only needs to be computed in the part of the computational domain which corresponds to the material. The results of g_0 are shown in Fig. 7.

The position and approximate size of the defect is found in any presented configuration. We notice that the addition of noise to the simulated measurements almost does not degrade the result (Fig. 7(c)). This is a good point for the robustness of the method. The use of a linear array with a coarser sampling significantly degrades the accuracy of detection, but the result is still exploitable (Fig. 7(d)). We can imagine that the TDTG can be a dimensioning tool for linear arrays, providing information on the number of transducers and the sizes to use to reach a required precision on the position and shape of the defects.

7. Discussion—the time reversal analogy

We begin this discussion with two small paragraphs, one about the versatility of the method, the second about the link between the migration theory of Tarantola and the time domain topological gradient.

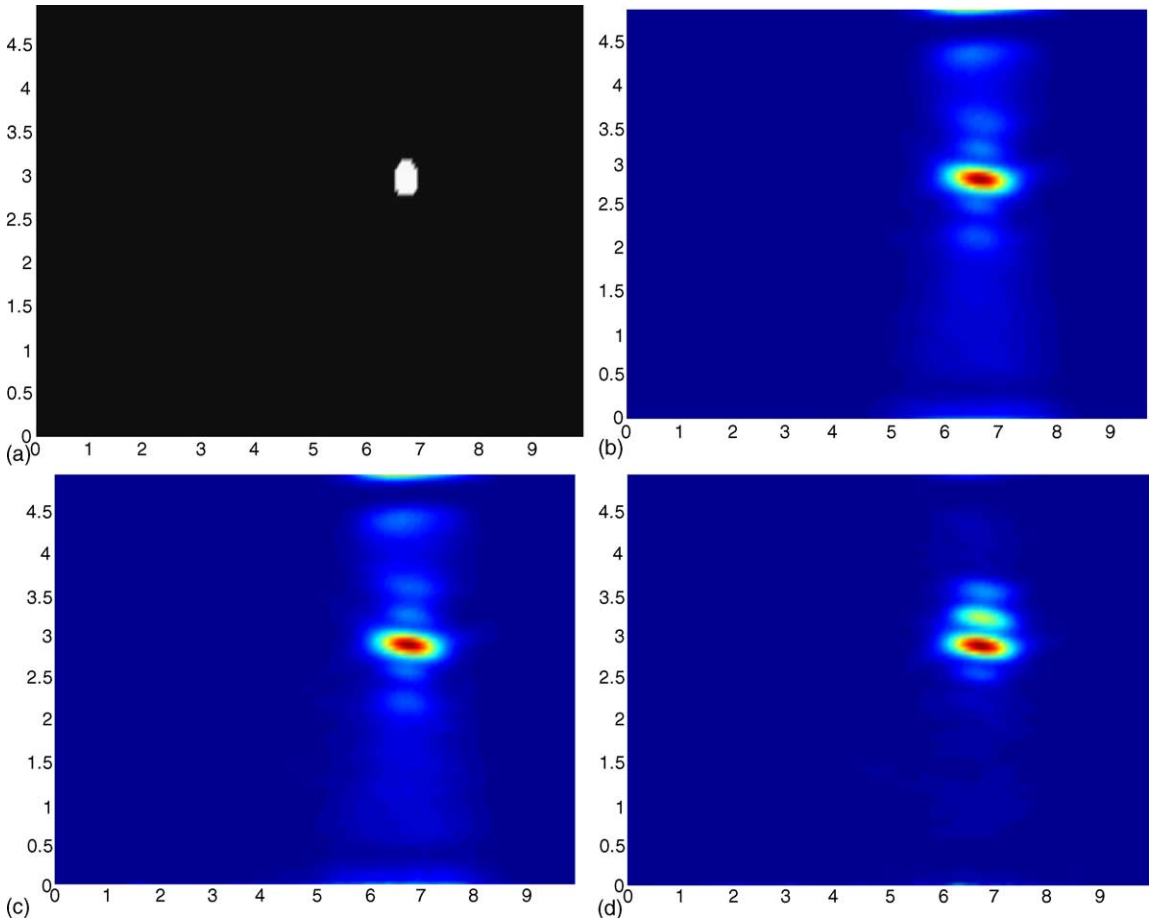

Fig. 7. Imaging of a defect in an immersed slab: (a) position of the defect in the slab, (b) isolevels of the topological energy g_0 in the slab with 36 transducers and no noise, (c) with 36 transducers and 5% of noise and (d) with 18 transducers and 5% of noise.

Then we will discuss in further details the analogy of the TDTG method with time reversal and its consequences.

7.1. Versatility of the TDTG method

It is worth noting that we can use quite small linear arrays of transducers, even a unique transducer can be enough. The counterpart is that we must scan the surface of the slab to inspect. This implies to solve the adjoint problem and compute a topological gradient for every position of the array, the final topological gradient being the average of the topological gradients of each position. Thus the method is really versatile, in the sense that one can do the measurements with the available ultrasonic apparatus, provided one is ready to pay the time computation price.

7.2. TDTG and migration

The migration method imagined by Tarantola [10] is fundamentally an imaging method of soils. It is a parameter optimization process, each iteration involving the computation of both a forward and an adjoint problem. The

topological gradient is basically a shape optimization method. When used in structural optimization with a full block of matter as the initial design and the Von Mises criterion as the cost function to optimize, it yields results very close to empirical optimal shapes [25]. It is particularly interesting to see that in the extent of the application to target detection, the topological gradient, through the choice of the cost function

$$j(\Omega) = \frac{1}{2}\int_{\mathbb{R}} \left(\int_{\Gamma_m} |u_\Omega - u_m|^2 \, \mathrm{d}x \right) \mathrm{d}t,$$

meets the principle of the migration method in the sense that we solve the same forward and adjoint problems. Indeed this is the differentiation of this cost function that leads to the source term $(u_m - u_0)$ in the adjoint problem. However the gradient of Tarantola is a classical gradient that gives a descent direction for the optimization of continuous parameters that are defined on the whole domain while the topological gradient allows a real modification of the topology of the domain. In other words, Tarantola's method allows one to deal with elastic heterogeneities while our method is also able to deal with very abrupt discontinuities such as holes or rigid inclusions. Moreover our method can give satisfactory imaging results at the end of the first iteration and therefore be far less time consuming.

7.3. The time reversal analogy

Traditional mathematical optimization algorithms, when used for physical applications, must usually be guided to converge to a physically realistic solution. The issue is then 'how to put physics in the method'. On the contrary, the time domain topological gradient method (as well as the migration method of Tarantola [10]) is intrinsically linked to physics. Indeed we recall the result of Section 5.1: in the time domain, the solving of the adjoint problem is a time reversal operation with the residuals $(u_m - u_0)$ acting as sources on the measurement zone Γ_m.

We will see that this result has great consequences on the method due to the physical properties of time reversal of ultrasonic waves, but first let us see through the example of Fig. 8 why the topological gradient reaches its target.

To understand what is happening during the time integration for the topological gradient computation, let us consider in parallel the forward and adjoint states. We show in Fig. 8 successive snapshots of the forward (left column) and adjoint (right column) displacement fields. In this example, only 16 of the 55 transducers are used in emission for the measurement and the forward problem, all of them work in reception and in emission for the adjoint problem. The contour of the unique defect is depicted (just for information) on the top-left picture of Fig. 8. As expected, the adjoint film runs backwards since it is solved from T to 0 and the result is put back from 0 to T for the needs of the time integration operation. The fields intercept each other at $t = t_{\text{int}}$ at the defect location, then we clearly guess that the scalar product of the two fields will be maximum for this intersection time $t = t_{\text{int}}$, and almost zero for the other times, yielding high amplitude values of the topological gradient at the defect position. We can also see on the right column that the compressional wave meets the shear wave of the adjoint problem at $t = t_{\text{int}}$ at the defect locus before they separate again. Only a small part of the shear wave travels here, the rest of it being not diffused in the array aperture during the measurement. Then it seems that the vectorial nature of elastic waves and the presence of various types of waves in the medium is not a problem towards the use of the TDTG method in elastic media. It is also worth noting that the method takes into account mode conversions since the chosen numerical solver accounts for them.

7.3.1. Consequences on the method

- The natural refocusing of the time reversed ultrasonic waves [5,26] added to the example of Fig. 8 contributes to the feeling that the TDTG method gives physical results. This pure algorithmic and mathematic method is then strengthened by the physics, which allows a certain confidence into the results. Using the optimization terminology we would say that the natural refocusing of time reversal prevents us from falling in a local minimum well which would lead even after numerous iterations to a non-physical solution. We can expect that, iterating, the method will converge towards a topology very close to the real one, provided enough measurement data are available (a

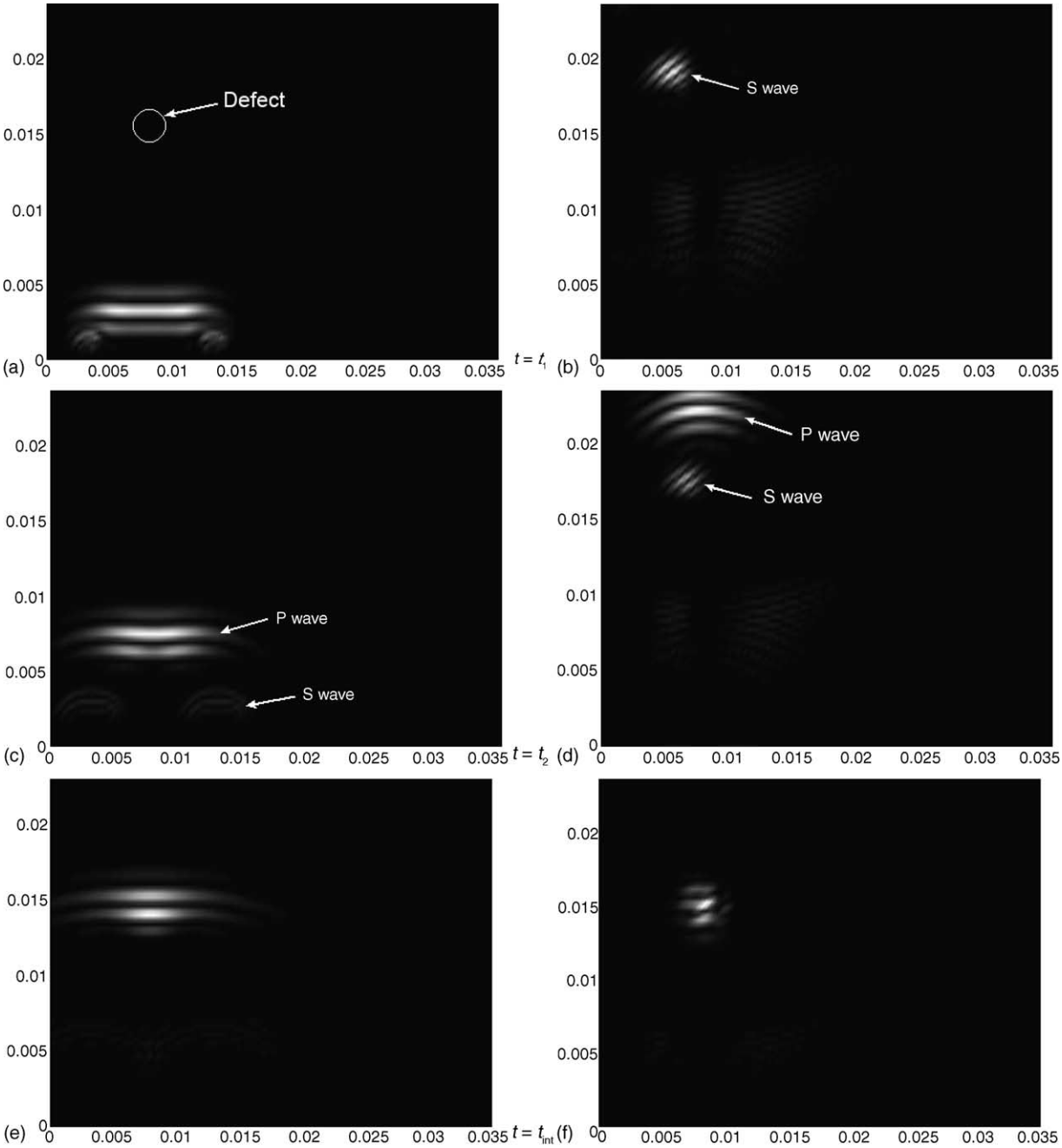

Fig. 8. Time integration process for the computation of the topological gradient. Forward displacement field in the reference medium without defect (left column), adjoint displacement field (right column). The fields intercept each other at $t = t_{\text{int}}$ at the defect location.

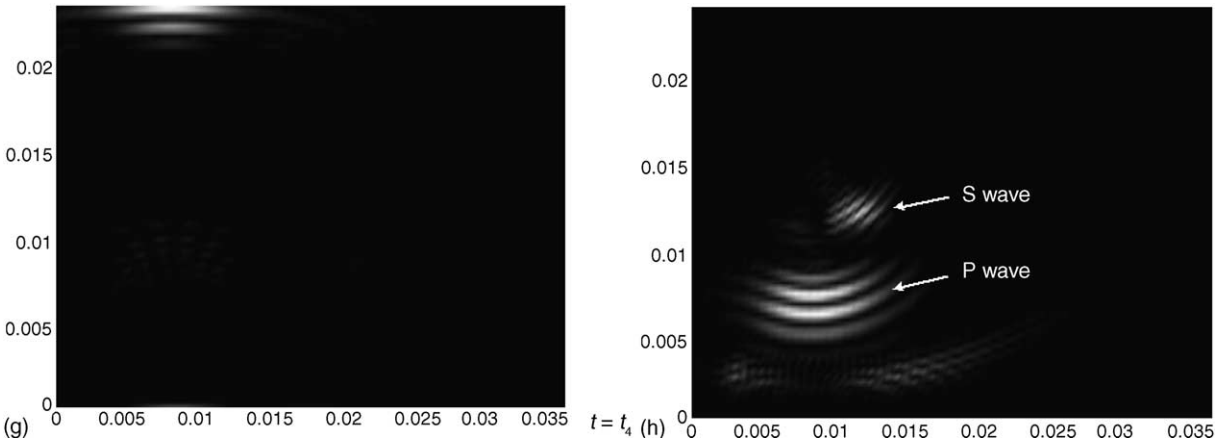

Fig. 8. (*Continued*).

measurement only on one side of the medium cannot give the full information required to perfectly reconstruct the medium (see Section 6.1)). Anyway what is noticeable in the TDTG is that it provides good imaging results without iterating the process. This makes it a very interesting method since it is very few time consuming and numerical instabilities due to the ill-posed nature of the inverse problem are avoided. The resolution limit of the method is given by parameters such as the limited aperture of the sensor array and its discretization sharpness. Moreover the focusing of the time reversal operation cannot be perfect, even in the case of a time reversal cavity, due to the absence of acoustic well at the defect location. At best, a focal spot of size $\lambda/2$ is observed [5] (see Section 6.1).

- As for the application of the method to lossy media in which time reversal invariance is broken, Tanter et al. [27,28] have shown that under the less restrictive assumptions of linearity and reciprocity, the time reversal process behaves as a spatio-temporal matched filter. That means that the refocused energy is maximum at the diffusion point and time, the degradation due to the lossy medium being a spreading of the energy around that maximum point. This allows us to expect that the method is robust and works in viscoelastic media despite a natural spreading of resolution.
- It is worth noting that the iterative method presented in Sections 5.2 and 6.1 has some similarities with the method proposed by Montaldo et al. [29] to reconstruct the inverse filter of the propagation with an iterative time reversal process based on successive corrections of the time reversed signal. Indeed the proposed iterative fixed point method of Section 5.2 acts, to some extent, the same way with a time reversed signal (the source of the adjoint problem) corrected iteration after iteration until convergence. As a result it is interesting to notice that our iterative method behaves like the heuristical approach of Montaldo et al. [30], detecting the strongest scatterer first and then the following in the decreasing order of scattered energy.
- Furthermore, the work of Tourin [31] have brought increasing knowledge and understanding of the physics of time reversal. The peculiar property of super-resolution have been studied on a mathematical point of view by Blomgren et al. [7]. This property is of particular interest in many application fields that involve imaging or focusing in random media. In non-destructive investigations we can imagine that sharp refocusing through multiple scattering media could be possible provided the intrinsic attenuation is small enough to assume the reciprocity of the medium. As a consequence we can expect, being very careful with it, that the TDTG method inherits this property and allows one to sharply image in media such as fiber-reinforced composites or grain steels. Indeed the repartition of scatterers in these media is random and the studies of [6,7] showed that, to some extent, the fact to use sources of short bursts in the time domain (thus wideband in the frequency domain) is a self-averaging process over the frequencies that propagate independently, and al-

lows one to consider a unique experience as an average over experiences performed on a great number of realizations of the random medium. Then the imaging would not be degraded by the change of statistical realization of the medium. That is to say we can consider the reference medium as one realization of the random medium and inspect a slab with another realization (with the same statistical properties) for imaging purpose.

This latter aspect, together with the taking into account of experimental data is our main prospect of research for the near future.

8. Conclusion

We have presented a topological gradient method in the time domain allowing one to find the accurate position and shape of obstacles, on the insonified side, in fluid or solid elastic media. A derived quantity has also been presented which allows one to obtain a good estimation of the global topology of the medium without iterating the process. Physical phenomena like mode conversions and multiple scattering are taken into account since the chosen numerical solver accounts for them. The method involves a cost function to minimize in order to converge towards the topology of the inspected medium. This cost function depends on the topology of the domain via the solution of the equations of elastodynamics in this domain. The first order terms of the asymptotic expansion of the cost function about the reference medium give the expression of the topological gradient whose more negative values indicate the presence of defects. The computation of the gradient requires the solving of a forward and an adjoint problem. We have seen that the adjoint problem corresponds to a time reversal operation. This confers a certain confidence to the results found with the TDTG method and recalls the back-propagation step of the migration method of Tarantola. We have also discussed the opportunity for the method to inherit special physical properties of the time reversal phenomenon. The method is relatively memory expensive but the first iteration usually gives satisfying results so it is very few time consuming. Besides the use of the method in the time domain with short pulse sources makes it particularly close to non-destructive testing or medical imaging conditions.

Acknowledgements

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Appendix A

The aim of this appendix is to show why we need to solve an adjoint problem in the computation of the topological gradient g and to find the expression of this problem. To do so let us have a look on how to write the variation of the cost function with respect to a change of the topology of the medium. For notations simplicity, let us consider the time-harmonic problem of the classical wave equation in 1D. It is obtained by considering solutions of the form $u(x)\,\mathrm{e}^{\mathrm{i}\omega t}$, which yields the equation

$$\Delta u + k^2 u = f \quad \text{with } k = \frac{\omega}{c}.$$

Taking, for instance and without loss of generality, a Dirichlet condition on Γ the external frontier of Ω_0, the problem becomes

$$\begin{cases} Au = f \text{ in } \Omega_0, \text{ with } A = \Delta + k^2 \\ u = 0 \quad \text{on } \Gamma \end{cases} \tag{A.1}$$

The multiplication of the previous equation by a regular test function $w \in \mathcal{V}$ and the integration by parts of the result (taking into account the boundary condition on Γ) gives the so-called variational (or weak) form of the problem (A.1)

$$\text{Find } u \in \mathcal{V} \text{ such that } a_0(u, w) = l(w) \quad \forall w \in \mathcal{V}, \tag{A.2}$$

where

$$a_0(u, w) = \int_\Omega (-\nabla u \cdot \nabla w + k^2 u v)\,\mathrm{d}x, \qquad l(w) = \int_\Omega f w\,\mathrm{d}x \tag{A.3}$$

and $\mathcal{V}$ is a Hilbert space having sufficient regularity properties so that (A.2) is well posed.

The insertion of a hole ω_ϵ in the domain Ω, with a Dirichlet boundary condition at its edge, gives the following problem:

$$\begin{cases} \Delta u_\epsilon + k^2 u_\epsilon = f & \text{in } \Omega_\epsilon = \Omega_0 \backslash \overline{\omega}_\epsilon \\ u_\epsilon = 0 & \text{on } \Gamma \\ u_\epsilon = 0 & \text{on } \partial\omega_\epsilon \end{cases} \tag{A.4}$$

and the associated weak formulation

$$\text{Find } u \in \mathcal{V} \text{ such that } a_\epsilon(u, w) = l(w) \quad \forall w \in \mathcal{V}. \tag{A.5}$$

Let u_0 and u_ϵ be the respective solutions to the problems (A.2) and (A.5).

We introduce the following Lagrange operator

$$\mathcal{L}(\epsilon, u, v) = J(u) + a_\epsilon(u, v) - l(v) \tag{A.6}$$

and

$$\mathcal{L}(0, u, v) = J(u) + a_0(u, v) - l(v) \tag{A.7}$$

which corresponds to the optimality criterion under the constraint that u verifies (A.5) (resp. (A.2)). It is clear that for all ϵ, $j(\Omega_\epsilon) = \mathcal{L}(\epsilon, u_\epsilon, v)$, including for $\epsilon = 0$. Therefore, for any $v \in \mathcal{V}$,

$$j(\Omega_\epsilon) - j(\Omega_0) = J(u_\epsilon) + a_\epsilon(u_\epsilon, v) - l(v) - J(u_0) - a_0(u_0, v) + l(v)$$

thus using a Taylor expansion at the first order for J, we obtain

$$j(\Omega_\epsilon) - j(\Omega_0) = a_\epsilon(u_\epsilon, v) - a_0(u_0, v) + D_u J(u_0)(u_\epsilon - u_0) + \mathrm{o}(f(\epsilon)), \tag{A.8}$$

where $D_u J$ designates the derivative of J with respect to the variable u.

Since the form $a(\cdot, \cdot)$ is bilinear, Eq. (A.8) can be rewritten as

$$j(\Omega_\epsilon) - j(\Omega_0) = a_\epsilon(u_\epsilon, v) - a_0(u_\epsilon, v) - a_0(u_0 - u_\epsilon, v) + D_u J(u_0)(u_\epsilon - u_0) + \mathrm{o}(f(\epsilon)). \tag{A.9}$$

Then choosing $v = v_0$ such that v_0 is a solution of

$$\text{Find } v \in \mathcal{V} \text{ such that } a_0(u_\epsilon - u_0, v) = -D_u J(u_0)(u_\epsilon - u_0) \quad \forall (u_\epsilon - u_0) \in \mathcal{V} \tag{A.10}$$

which rewrites

$$a_0^*(v, u_\epsilon - u_0) = -D_u J(u_0)(u_\epsilon - u_0) \tag{A.11}$$

and admitting that $\|a_\epsilon(u_\epsilon, v_0) - a_0(u_\epsilon, v_0) - f(\epsilon)g(u_0, v_0)\| = \mathrm{o}(f(\epsilon))$, we obtain the asymptotic expansion (3):

$$j(\Omega_\epsilon) = j(\Omega) + f(\epsilon)g(u_0, v_0) + \mathrm{o}(f(\epsilon)).$$

The associated strong formulation of the problem (A.11) is

$$A^* v = -D_u J(u_0). \tag{A.12}$$

The solution v_0 of (A.12) is called the adjoint state since A^* is the adjoint operator of A. Then the computation of the topological gradient (i.e. the first order terms of the asymptotic expansion of j) involves the solving of the adjoint problem (A.12) and of the associated forward problem (A.1).

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